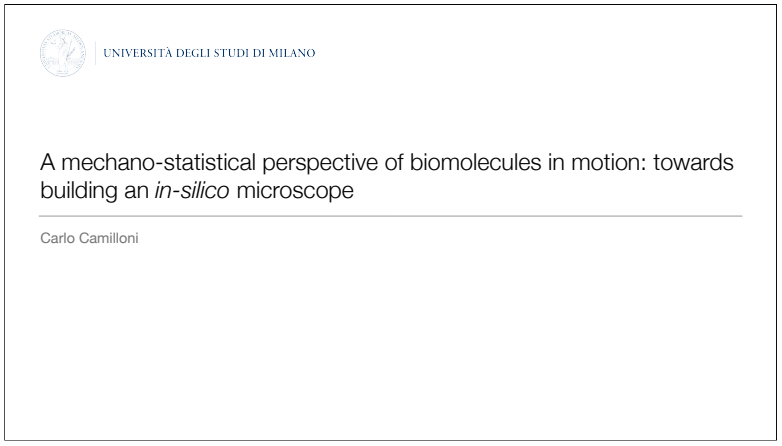
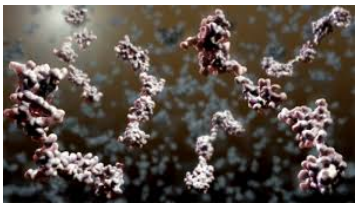


This lecture builds, step by step, a statistical-mechanical language for describing biomolecules in motion. The long-term goal is to use this language to interpret and later build the computational models used throughout the rest of the course (the ‘in-silico microscope’).

Why a Statistical and Mechanical perspective

When we observe/describe biomolecules we generally handle many molecules, made of many atoms, that can:

- change their shape;
- move in 3D space;
- have interactions;



To describe all these we need to know how they move and interact (mechanics) and how to handle large numbers (statistics)

Main idea: why do we need both statistics and mechanics to describe biomolecules?
A molecular system is made of many atoms that are constantly changing shape, moving in 3D, and interacting with each other and their environment. A single, instantaneous snapshot is almost never representative on its own.
Statistics gives us tools to summarise what happens over many atoms/many observations; mechanics gives us the physical rules (forces, energy) that make some arrangements more likely than others. Combining the two lets us connect what we can observe (populations, frequencies) to what is physically happening (energies, forces).

Why a Statistical and Mechanical perspective

In the next I want to show you how statistical observations can be interpreted using the language of mechanics: energy.



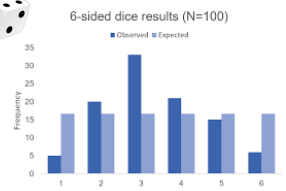
Probabilities are estimated using histograms: the problem of sampling relative and absolute probabilities



Stochastic events are estimated by SAMPLING. Sampling means exploring a probability distribution looking at many outcomes organised as histograms.

Absolute probability needs to have information on all outcomes, relative probabilities are just frequency ratios:





Absolute probability: p(i) = count(i)/(sum_k(count(k)))
Relative probability: p(j)/p(i) = count(j)/count(i)

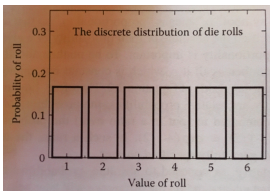
Main idea: how do we actually estimate a probability in practice?
We never observe a probability distribution directly — we SAMPLE it, by generating many outcomes and organising them into a histogram.
Two different quantities can be computed from a histogram: the ABSOLUTE probability of outcome i, p(i) = count(i)/sum_k(count(k)), which needs the full histogram over all outcomes; and the RELATIVE probability between two outcomes, p(j)/p(i) = count(j)/count(i), which only needs their two counts.
Remember: relative probabilities require much less sampling than absolute probabilities, which is important in practice whenever full sampling is expensive.

A stochastic picture for molecules in motion: probabilities

Stochastic means randomly determined, events happen by chance, by observing many events one can learn **what is the probability of the different outcomes** that is one can learn their **probability distribution**.

If some outcome is more likely than another one could think in terms of energy that that outcome is favourable in **energy**.





p(j)=1/6 with j=1 to 6 **Discrete**

$\sum_{j=1}^{6} p(j) = 1$ Normalised
When all possibilities are accounted for the probability is 100%

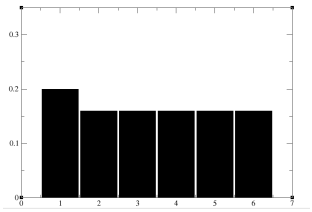
Main idea: what does ‘stochastic’ mean, and how do we represent a probability distribution, using a familiar example (a die)?
Stochastic means randomly determined: by repeating an experiment (rolling a die) many times we can estimate how likely each outcome is, i.e. its probability distribution. If one outcome is systematically favoured, we can already start to think of this in energy terms: that outcome could be said to be energetically more favourable.
A fair six-sided die gives a flat, discrete distribution, p(j) = 1/6 for every face j. It is normalised: summing over all 6 possible outcomes gives exactly 1 (100%).
Remember: any complete probability distribution must sum (or integrate) to 1 once every possible outcome is included.

A stochastic picture for molecules in motion: biases

MUST KNOW

If your dice is biased towards a given number, let's say 1, then your histogram will be higher at 1. In terms of energy you can think that the state 1 is energetically more favourable.





$$\sum_{j=1}^{6} p(j) = 1$$

Normalised
When all possibilities are accounted for the probability is 100%

To bias a dice you need to change some of its mechanical properties, that generally speaking will have to do with it being thrown under the influence of gravity

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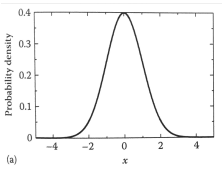
Main idea: what happens if the die is biased, and how do we interpret that physically?
If a die favours one face (say 1), its histogram will peak there. We can describe this in energy language: state 1 is energetically more favourable than the others, the same way a ball tends to settle at the bottom of a valley.
Practically, biasing a real die means changing a MECHANICAL property of the die (e.g. shifting its centre of mass), since it's the die's behaviour under gravity — not an abstract probability — that actually sets the outcome frequencies. This is the first concrete bridge between statistics (a skewed histogram) and mechanics (an unevenly distributed mass, acted on by gravity).

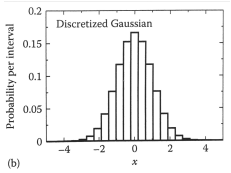
Continuous Probability Distributions are also estimated using histograms

$$\rho_G(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-<x>)^2}{2\sigma^2}}$$

Gaussian

Even if we know only proportionality this can be enough to make comparisons: the probability of p(0)/p(1). This is clear if you think about histograms.





To calculate a probability one multiplies for a small interval.

Density, they are in units^{-1}

Main idea: extending histograms to CONTINUOUS variables — probability density functions.
For continuous quantities we still build histograms, but the relevant object is now a probability DENSITY (units of 1/x, where x is whatever we're measuring), the Gaussian being the classic example.
Densities are only meaningful together with a (small) interval: to get an actual probability from a density you multiply the density value by the width of that interval.
As in the discrete case, even knowing only a density up to a proportionality constant is enough to compute ratios like p(0)/p(1), which is often all that's needed.

Average, Standard Deviation, Standard Error of the Mean



In practice we **estimate** averages using sums:

$$< f > \doteq \frac{1}{N} \sum_{i=1}^{N} f(x_i)$$

Variance and Standard deviation:

$$\sigma^2 = var(f) = < (f - < f >)^2 > = \int dx (f(x) - < f >) \rho(x) \doteq \frac{1}{N-1} \sum_{i=1}^{N} (f(x_i) - < f >)^2$$

Tell us about the width of the distribution $\rho(x)$.

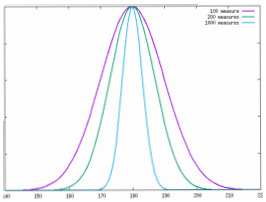
Main idea: how do we summarise a distribution with a single number (the average), and describe its spread (standard deviation / variance)?
With a finite sample, we estimate the average as a simple sum (mean of the observed values) and the spread as the variance (mean squared deviation from that average) or its square root, the standard deviation.
Remember: the standard deviation describes the width of the underlying distribution ρ(x) itself — how broadly individual observations are scattered — NOT how precisely we know the average. That distinction is the subject of the next slide.

Standard Error of the Mean



Standard error: $\text{Std-err} = \sqrt{\frac{var(f)}{N}} = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1}{N^2 - N}\sum_{i=1}^{N}(f(x_i) - < f >)^2}$

Tell us about **accuracy of our estimate of the average,** the idea is that our average value that we estimate from multiple observations has a resolution with the shape of a Gaussian with the width of the standard error.



Main idea: how precisely do we actually know the average we computed? This is a different question from the previous slide's spread of individual values.
The standard error of the mean tells us about the ACCURACY of our estimate of the average: if we repeated the whole experiment (collecting a new set of observations) many times, the estimated averages would themselves be scattered, following a Gaussian distribution whose width is the standard error.
Remember: standard deviation describes the individual data; standard error describes how reliable the AVERAGE of that data is. Mixing up the two is one of the most common mistakes when reporting results.

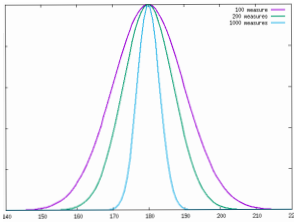
Average, Standard Deviation, Standard Error of the Mean



$$< f > \doteq \frac{1}{N}\sum_{i=1}^{N} f(x_i) \qquad \sigma^2 = var(f) = < (f - < f >)^2 > = \int dx (f(x) - < f >) \rho(x) \doteq \frac{1}{N-1}\sum_{i=1}^{N} (f(x_i) - < f >)^2$$

$$\text{Std-err} = \sqrt{\frac{var(f)}{N}} = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1}{N^2 - N}\sum_{i=1}^{N} (f(x_i) - < f >)^2}$$

The average of a random process is always distributed as a gaussian of width std-err and so you can set confidence intervals: (±1std-err is 68%, ±2std-err 95%, ±3std-err 99%, …). This is called the theorem of the central limit.



Main idea: confidence intervals, from the Central Limit Theorem.
Because the average of a random process is always distributed as a Gaussian — regardless of the shape of the original distribution being averaged, this is the Central Limit Theorem — we can use the standard error to build confidence intervals around an estimated average: roughly 68% confidence within ±1 standard error, 95% within ±2, 99% within ±3.
This is an extremely common rule of thumb any time you report an average computed from a finite sample.

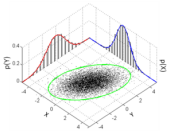
Multidimensional distribution function: projection and correlation

Probability distributions can be more than 1D. For example the probability distribution of a molecule around a receptor may be studied in terms of (x,y,z) coordinates of the center-of-mass, in this case would be 3D.

$\int \rho(x,y)dxdy = 1$ Normalisation

How can we show a multi dimensional distribution in less dimensions? By projecting it:

$\rho(x) = \int \rho(x,y)dy$ Projection/Marginalisation



$$< x >= \int dxdyx\rho(x,y)$$

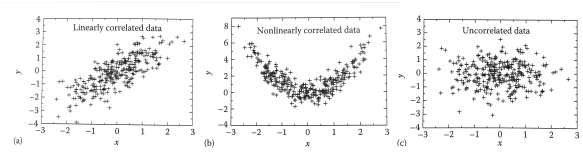
$$< y >= \int dxdyy\rho(x,y)$$

$$< xy > \int dxdyxy\rho(x,y)$$

By looking at the data can you think of a projection that is more important to understand the data?

Main idea: distributions can live in more than one dimension — how do we visualise and interpret them?
A probability distribution can depend on several variables at once: for example, the (x,y,z) position of a molecule's centre of mass around a receptor is a 3D distribution. Since we can't easily visualise more than 2–3 dimensions at once, we often PROJECT (marginalise) a multidimensional distribution onto fewer dimensions, by integrating/summing out the variables we're not interested in. The art is choosing a projection that keeps the physically or biologically relevant information while discarding the rest.

Multidimensional distribution function: correlation



$$< x >= \int dxdyx\rho(x,y)$$

$$< y >= \int dxdyy\rho(x,y)$$

$$< xy > \int dxdyxy\rho(x,y)$$

If two variables are independent than

$$\rho(x,y) = \rho(x)\rho(y)$$ **So**

$$< xy >=< x >< y >$$

So we have correlation if

$$< xy > - < x >< y >\neq 0$$

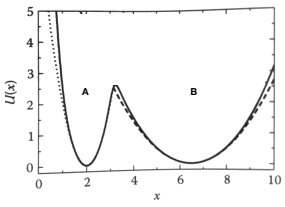
Pearson correlation coefficient: $$r = \frac{< xy > - < x >< y >}{\sigma_x \sigma_y}$$

Main idea: when are two variables correlated, and how do we quantify it?
If two variables were completely independent, their joint probability would simply be the product of their individual (marginal) probabilities.
When this is not the case, the variables are correlated. The Pearson correlation coefficient gives a standard, normalised way (between –1 and +1) to quantify how strongly — and in what direction — two variables move together.

Statistical Mechanics: the Boltzmann distribution

What is the connection between statistics and mechanics?

That is the Boltzmann equation: $pdf(x,v) \equiv \rho(x,v) \propto \exp\left[\frac{-E(x,v)}{k_B T}\right] = \exp\left[\frac{-U(x)}{k_B T}\right] \exp\left[\frac{-K(v)}{k_B T}\right]$



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Main idea: what is the precise mathematical link between mechanics (energy) and statistics (probability)? This is the Boltzmann distribution, and it's the central concept of the whole lecture.
Up to now we've only talked qualitatively about low-energy states being more probable. The Boltzmann distribution makes this quantitative: the probability of observing a molecule in a given conformation depends exponentially on that conformation's energy (and on the temperature).
Every later example in this lecture — ligand binding, protein folding — is really just an application of this one equation.

gives the probability to observe a molecule in a given conformation



Let's look at this pdf in more detail: $pdf(x) \equiv \rho(x) \propto \exp\left[\frac{-U(x)}{k_B T}\right]$

$$pdf(v) \equiv \rho(v) \propto \exp\left[\frac{-K(v)}{k_B T}\right] = \exp\left[\frac{-\frac{1}{2}mv^2}{k_B T}\right]$$

Conformations and velocities are independent. This means that we can study them separately. The distribution of the velocities does not affect the distribution of the configurations. Furthermore the distribution of the velocity is Gaussian, so it is easy to integrate it.

So for all practical purposes we can just ignore the velocity: $pdf(x) \equiv \rho(x) \propto \exp\left[\frac{-U(x)}{k_B T}\right]$

This gives a link between the energy of a conformation and the probability of observing it

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Main idea: why can we focus only on a molecule's configuration (its shape) and ignore its velocities?
The full distribution predicted by statistical mechanics depends on both conformation and velocities, but these factor apart: the velocity distribution is independent of the conformation (it's always Gaussian, tied only to temperature), so it doesn't affect which conformations are favoured.
In practice this means we can integrate the velocity part away and are left with a distribution over conformations only, where a conformation's energy directly sets how probable it is.
Remember: p(conformation) ∝ exp(–E/kT) — lower energy means higher probability, and temperature sets how sensitive this is to energy differences. This is the practical, usable form of the Boltzmann distribution you will use for the rest of the course.

Statistical Mechanics: Ergodicity



1. **A single molecule (at constant temperature) will move through all the possible conformations at random due to the collisions with the solvent. Each conformation is gonna be populated accordingly to the energy and temperature (the interaction with environment).**
2. **If at a given time one stops the movie of many many copies of the same system in the same condition the picture taken will display conformations according to the same statistics.**

The collection of conformations in both cases (that is the sampling) is usually called an ensemble of conformations. This ensemble is characterised first of all by the conditions in which it has been acquired (constant temperature, pressure or volume, constant number of particles, ecc)

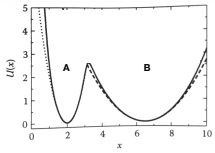


Main idea: ergodicity — why can a single molecule, observed over time, tell us about a whole population?
A single molecule at constant temperature will, due to random collisions with the solvent, visit all accessible conformations over time, spending more time in low-energy ones (consistent with Boltzmann). If instead you took a snapshot of many independent copies of the same system under the same conditions, you would see the same statistics. This equivalence between ‘one system observed for a long time’ and ‘many systems observed at one instant’ is called ergodicity. The resulting collection of sampled conformations, however it was generated, is called an ensemble — always characterised by the conditions under which it was acquired (temperature, pressure, volume, number of particles, etc).

Free Energy: States and Probabilities

State: is a collection of configurations (microstate), ideally belonging to the same potential energy basin.

A simple 1D potential energy



Here we can visually define two states A and B for the two basins, e.g. all the conformations belonging to A and to B

$$p_A = \int_{V_A} \rho(x)dx \propto \int_{V_A} \exp[-U(x)/k_BT]dx$$

$$p_B = \int_{V_B} \rho(x)dx \propto \int_{V_B} \exp[-U(x)/k_BT]dx$$

The Free-Energy is the "effective" energy of a state, i.e. the energy that will give the same probability

$$\frac{p_A}{p_B} = \frac{\int_{V_A} \exp[-U(x)/k_BT]dx}{\int_{V_B} \exp[-U(x)/k_BT]dx} \equiv \frac{\exp(-F_A/k_BT)}{\exp(-F_B/k_BT)} \qquad F_i \propto -k_BT \ln\left(\int_{V_i} \exp[-U(x)/k_BT]dx\right)$$

Main idea: grouping many microscopic conformations into macroscopic STATES, and defining their free energy.
Rather than tracking every single conformation, we often group together all conformations belonging to the same energy basin (e.g. basin A, basin B) into a single STATE. Since a state actually contains many microscopic conformations, its effective energy is not simply the energy of one configuration: it is the FREE ENERGY, defined as the single energy value that, if every conformation in that state had exactly that energy, would reproduce the same overall probability of observing the state.

Summary: relative probabilities, energy and time



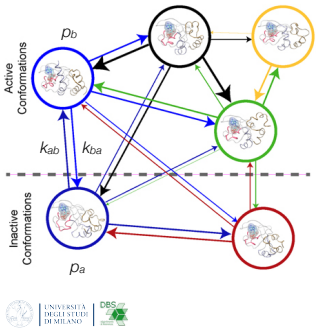
Microscopic processes are guided by **energy**, between two microscopic state their relative probability is associated exponentially to their energy difference.

Macroscopic processes (made of many microscopic one) are guided by **free-energy**, between two macroscopic state their relative probability is associated exponentially to their free-energy difference, the number of microscopic states making a macroscopic state is the entropy of the macroscopic state.

Macroscopic Processes happen in a given average time that is associated the free-energy barrier separating them

Main idea: summary — connecting probability, energy/free-energy, and time, at two different levels of description.
At the MICROSCOPIC level (individual conformations), the relative probability between two conformations scales exponentially with their energy difference.
At the MACROSCOPIC level (states made of many conformations), the relative probability between two states scales exponentially with their FREE-energy difference, and the number of microscopic conformations contained in a state is its entropy.
Finally, how long a macroscopic exchange between states takes is set by the FREE-ENERGY BARRIER separating them — not by the free-energy difference between the states themselves. This is where TIME/RATES enter the picture, as something distinct from population/probability.

Summary: A stochastic picture of molecules in solution



Macrostate populations are associated with free energy differences:

$$p_i \propto \exp\left(-\frac{G_i}{RT}\right)$$

Exchange rates are associated with free energy barriers:

$$k_{ij} = k_0 \exp\left(-\frac{G_{ij}^{\ddagger}}{RT}\right)$$

Energy for these processes is provided by thermal energy, so essentially by collision with water molecules

An equilibrium bulk experiment ideally capture the contribution from all the states:

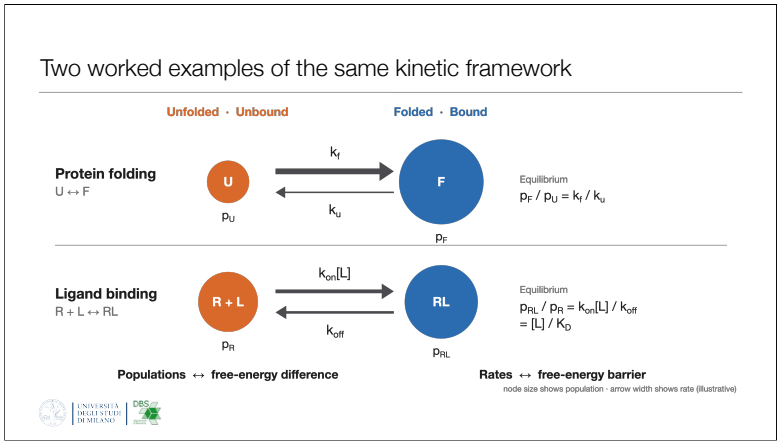
$$< O >= \sum_i p_i O_i = \frac{\sum_i O_i \exp\left(\frac{-G_i}{RT}\right)}{\sum_i \exp\left(\frac{-G_i}{RT}\right)}$$

Microscopically:

$$G_i = -RT \ln \int \exp\left(-\frac{U(r)}{RT}\right) \delta(i - i(r)) dr$$

Main idea: putting it together — populations and rates for a two-state system (A ⇌ B), and the connection to a real bulk experiment.

The relative populations pa, pb of two macrostates are set by their free-energy difference, while the exchange rates between them (kab, kba) are set by the free-energy barrier between them.
The thermal energy driving these transitions comes from random collisions with the solvent (water).
A bulk experiment at equilibrium samples contributions from ALL molecules/states simultaneously, so what you measure experimentally is an ensemble average over this microscopic picture of individual molecules hopping between states with rates kab/kba.



Main idea: we now apply exactly this probability-rate-energy framework (previous several slides) to two concrete molecular examples.
Both ligand binding and protein folding are modelled as two-state (or few-state) systems: bound vs unbound for binding, folded vs unfolded for folding. In both cases we describe the system with (1) the relative POPULATION of the two states, linked to their free-energy difference, and (2) the RATES of interconversion between them, linked to the free-energy barrier.
Seeing the same mathematical structure show up in two very different biological processes is the point of this part of the lecture.

Ligand binding: a molecular example

Here energy and probability are intuitively linked, furthermore, events happen on some **time scale**, so one would like to find a relationship between probability, energy and time scales of events. This and the next example (protein folding) both use exactly this probability-rate-energy framework, just applied to a different process.

$$R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$$

In a ligand-receptor binding process we can picture the process, at least in some cases, as a probability for R and L to be unbound vs a probability to be bound (RL) and consider the time scale of the process in terms of rates of binding (on) and unbinding (off).

The k_{off} rate is the frequency of unbinding (s^{-1}) so that $k_{off}[RL]$ is the number of unbinding events per second.

The k_{on} represent instead two processes (1) collision and (2) complexation. So this is the collision frequency times the complexation probability, this means that binding depends on concentration, indeed k_{on} is measured in (s^{-1} M^{-1}). $k_{on}[R][L]$ is the number of binding events per second, while $k_{on}[L]$ is the frequency of binding for a given concentration of ligand (excess of ligand).

Main idea: ligand binding as a two-state kinetic system.
We describe ligand(L)-receptor(R) binding in terms of a probability of being unbound (R+L) vs bound (RL), together with the TIME SCALE of the process, captured by an unbinding rate (koff) and a binding rate (kon).
koff is simply the frequency of unbinding events per second. kon is more subtle: it combines the frequency of COLLISIONS between R and L with the PROBABILITY that a collision actually leads to a successful complex (complexation). Because collisions depend on how much ligand/receptor is around, kon depends on concentration — this is why kon has different units (s-1 M-1) from koff (s-1).

A stochastic picture: ligand binding

$$R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$$

If we start from a solution of free [R] and [L] we start forming RL with rate $k_{on,}$ once we have some [RL] this can also dissociate with rate k_{off}, after some time we will be at equilibrium that is that the concentration of [R], [L] and [RL] will not change anymore.

$$K_d \equiv \frac{k_{off}}{k_{on}} = \frac{[R][L]}{[RL]} = C^0 \frac{p_{unbound}}{p_{bound}}$$

The **dissociation constant** is can then be defined as the ratio between k_{off} and $k_{on,}$ it is measured in (M), where lower values means stronger binding (i.e. lower concentrations of ligand are enough to bind)

$$C^0 = 1\ M$$

$$\Delta G_{\text{binding}} = k_{\text{B}} T \ln\left(\frac{K_{\text{d}}}{C^0}\right) = k_{\text{B}} T \ln\left(\frac{k_{\text{off}}}{k_{\text{on}} C^0}\right)$$

$$= k_B T \ln\left(\frac{p_{unbound}}{p_{bound}}\right)$$

This is the binding free-energy that is the amount of work that can be extracted or that must be provided for the binding reaction to happen, and you see is related to the rates as well as to the probabilities



Main idea: from rates to equilibrium, and to the binding free energy.
Starting from free R and L, RL forms at rate kon and can dissociate at rate koff; after enough time the system reaches equilibrium, where the concentrations of R, L and RL stop changing on average.
The ratio koff/kon defines the DISSOCIATION CONSTANT KD (units M): the lower KD, the stronger the binding (less ligand needed to bind).
KD is directly related to the BINDING FREE ENERGY, the work that must be supplied (or that can be extracted) for the binding reaction to occur — connecting the kinetic picture (rates) back to the thermodynamic picture (free energy) from earlier in the lecture.

A stochastic picture: protein folding (same framework as ligand binding)

$$U \underset{k_u}{\overset{k_f}{\rightleftharpoons}} F$$

Their ratio defines the $_K$D=$_k$u/$_k$f. This can be used to calculate the difference in free energy:

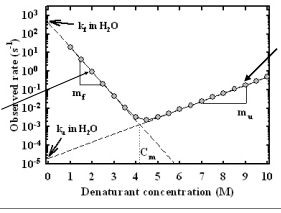
$$\Delta G_{Eq} = RT \ln K_D$$

The k_u rate is the frequency of unfolding (s^{-1}) so that k_u[P] is the number of unfolding events per second.

The k_f rate is the frequency of folding (s^{-1}) so that k_f[P] is the number of folding events per second.

The two rates can be measured by changing the concentration of denaturants by a stopped-flow and monitoring fluorescence.

Protein in high-conc of denaturant is mixed with buffer to a target final low concentration of denaturant



Protein in buffer is mixed with denaturant to a target final high concentration of denaturant

At high C all the protein unfold $k_{obs}=k_u$ at low C all the protein fold so $k_{obs}=k_f$

Main idea: protein folding, described with exactly the same framework as ligand binding.
Folding is modelled as a two-state process between an unfolded (U) and folded (F) conformation, with an unfolding rate ku and a folding rate kf, directly analogous to koff/kon for binding. Their ratio defines an equilibrium constant (called KD here by analogy), from which the folding free energy can be computed.
Experimentally these rates are measured with stopped-flow fluorescence: the protein is rapidly mixed either from high to low denaturant concentration (favouring folding — you measure kf) or from low to high denaturant concentration (favouring unfolding — you measure ku).
At very high denaturant concentration essentially all molecules unfold, so the observed relaxation rate kobs approaches ku; at very low concentration essentially all molecules fold, so kobs approaches kf.

The common theme of Structural Bioinformatics: sampling and energy (or score, or loss function)



In the next lectures we will see how the problem of SAMPLING and the problem of the ENERGY or SCORING FUNCTION or LOSS FUNCTION, are interconnected and always present. These are the problem to cope with when doing MD simulation, structure prediction, docking, protein design, …

You can think at the next lectures as to specific strategies to cope with these two problems when trying to address specific problems. All the algorithms we will see are related to them (MD, MC, EM, ML/AI, …)

Main idea: bridging to the rest of the course — sampling and energy/scoring/loss are the two recurring problems behind every method you will see.
Every computational method in structural bioinformatics — molecular dynamics, Monte Carlo, structure prediction, docking, protein design, machine learning — ultimately has to solve two intertwined problems: how to SAMPLE the relevant states efficiently, and how to define/evaluate an ENERGY (or score, or loss function) that tells you which states are good.
You can think of the rest of the course as a tour of different strategies for tackling these same two problems in different contexts.

Recap

- Microscopic probability ↔ energy: $p \propto \exp(-E/kT)$
- Macroscopic (free-energy) probability ↔ entropy and energy of a state
- Exchange rates ↔ free-energy barriers between states
- Ligand binding and protein folding: same rate framework, two different molecules

Main idea: recap of the lecture's core logical chain.
(1) At the microscopic level, probability and energy are linked exponentially through the Boltzmann distribution.
(2) At the macroscopic/state level, probability is linked to FREE energy, which bundles together the energy and the entropy (number of microstates) of a state.
(3) The TIME it takes to exchange between states is governed by the free-energy BARRIER separating them, not by their free-energy difference.
(4) Ligand binding and protein folding are two concrete molecular examples of exactly this same rate framework, just applied to two different physical processes.

Questions:

- How do you calculate the average outcome of a stocastic process?
- What is the difference between the standard deviation and the standard error of the mean?
- Which is the probability distribution function of the average of a process?
- What is the difference between the absolute probability of an event and the relative probability of that event with respect to another?
- What is the relationship between the probability of a macrostate and its free-energy?
- What is the relantionship between the probability of a configuration and its energy?
- What is the free energy change of a process?
- …

